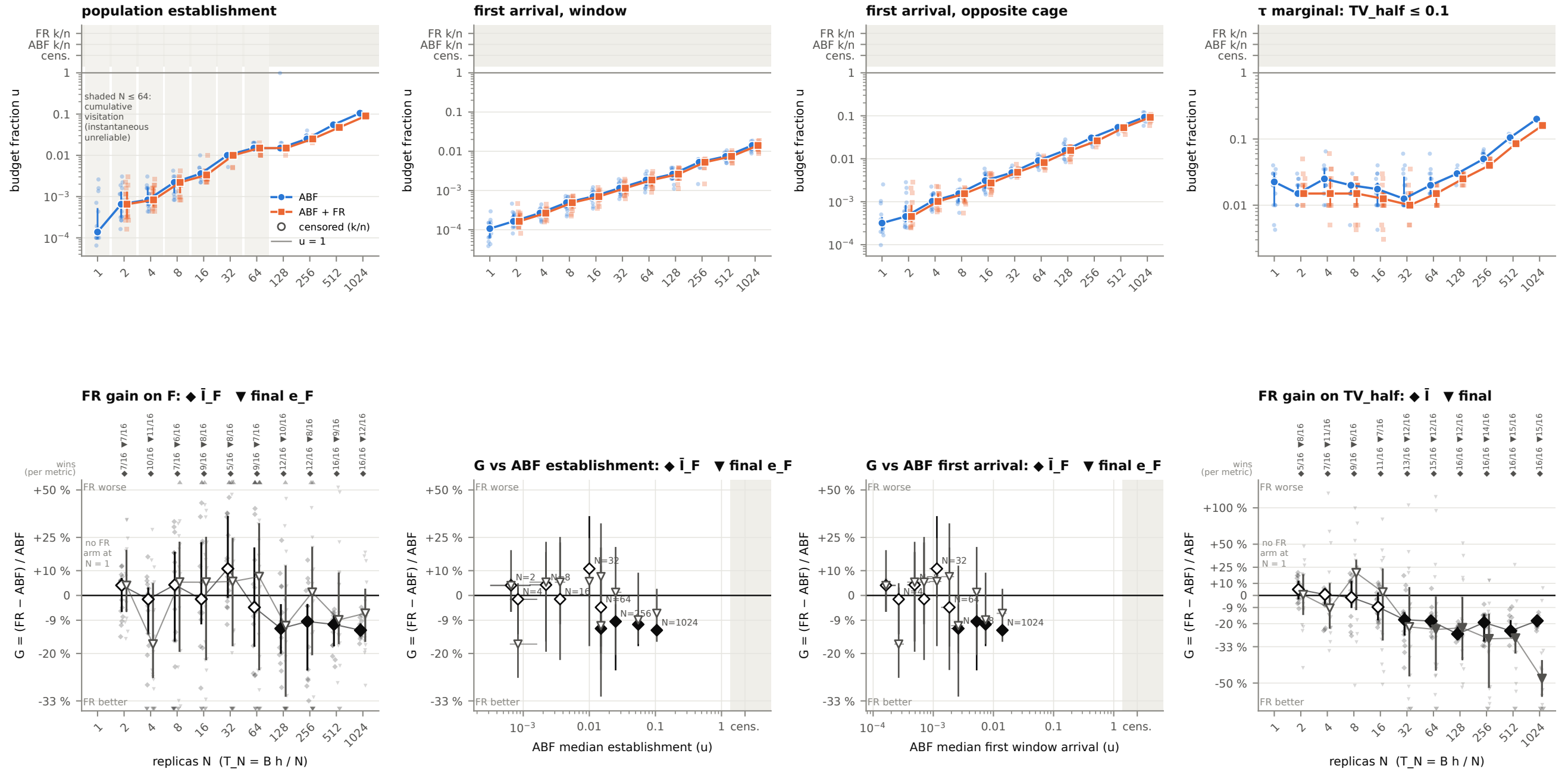


ethane/LTA 300 K: population establishment, first arrival and marginal convergence vs N (budget fraction)

Equal total budget per arm per seed: $B = N \times n_steps = 3.072e+08$ walker-steps at every N ($h = 0.0002$, $T_N = B \cdot h / N$); 16 seeds per (N, method). Do FR gains coincide with establishment-limited N?



ethane/LTA 300 K · S6a_establishment_budget · status: all planned (N, seed, method) complete · FR is not defined at N = 1 (ABF only, by design) · LTA N < 50: FR uses the finite-N cap extension $\max(1, \text{floor}(0.02 N))$. Establishment = first save after which the instantaneous far-state fraction (window $|z| < 1.5$ A) stays $\geq \frac{1}{2} \times 0.252$. Shaded N [1, 2, 4, 8, 16, 32, 64]: the instantaneous rule is UNRELIABLE there (ill-defined at $N \leq 2$; an exactly uniform population passes every save of the second half with probability < 0.9 , summary establishment_null), so the same rule on the cumulative visitation is plotted; FR birth-death can push the instantaneous count below binomial scatter, so an FR establishment advantage at those N is not evidence of faster convergence. $\tau(\text{TV_half})$ on the budget axis uses the common budget grid. Top: seed median (large open marker in the 'cens.' band = median censored), IQR bars, faint seeds. Bottom: paired G with bootstrap 95% CIs (filled = excludes 0; 'wins' = FR < ABF / n); scatters: each N's G against the ABF median time (horizontal bar = ABF IQR; 'cens.' = censored). Source results/equal_budget_v2/lta_T300/analysis/summary.json (generated 2026-10-10T07:06:16Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.